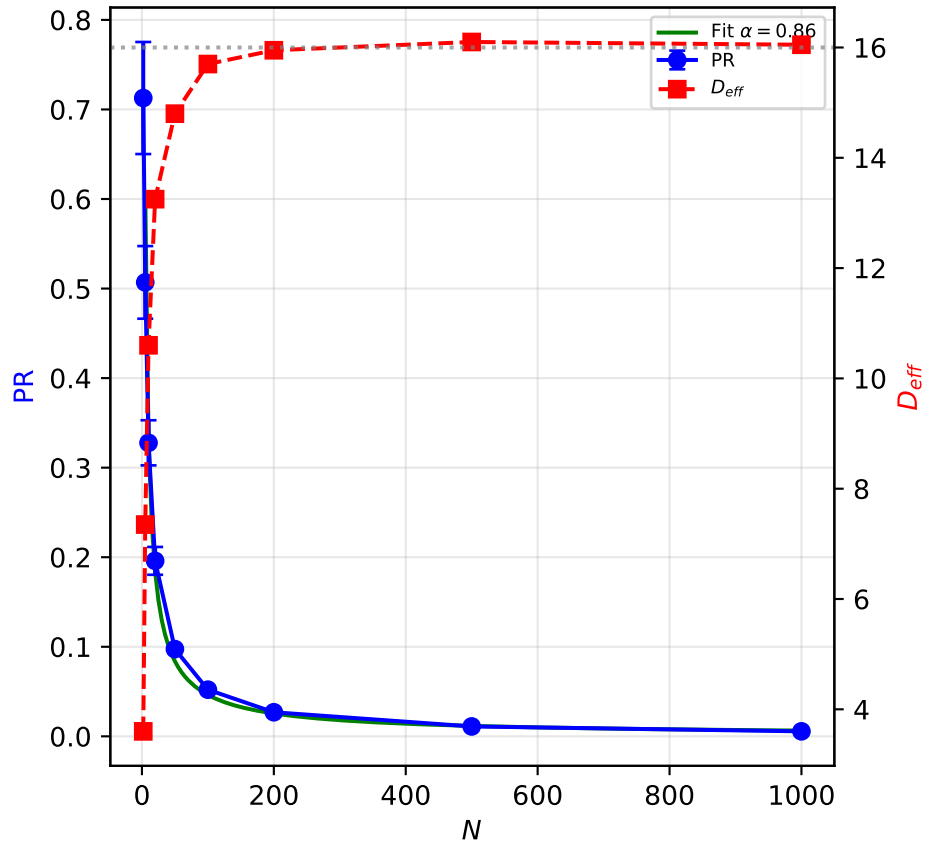
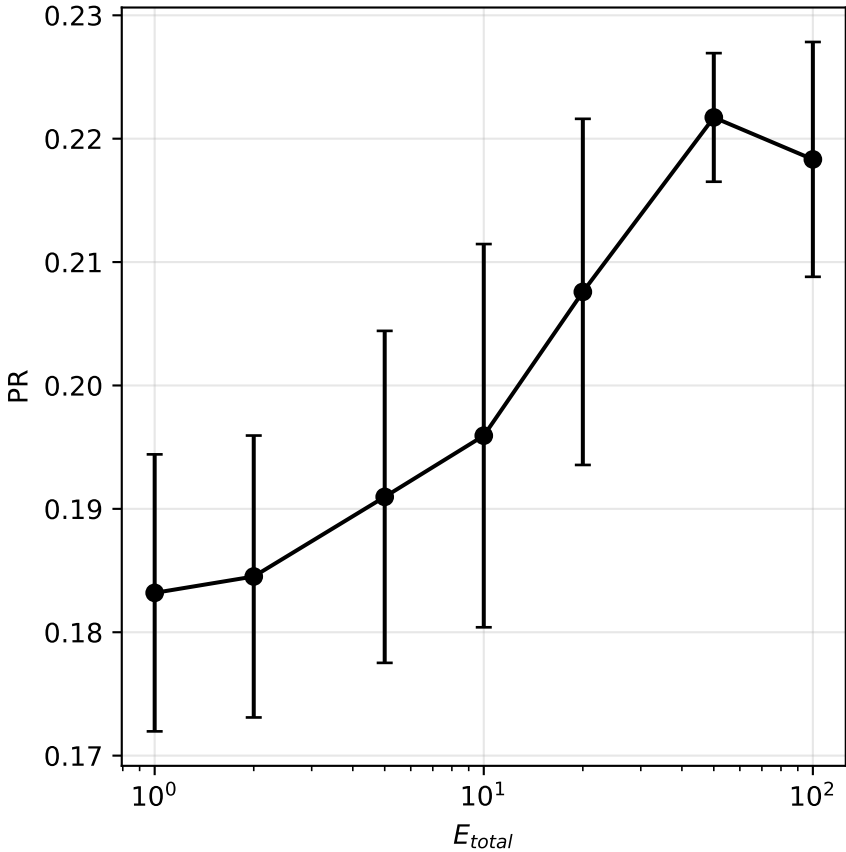


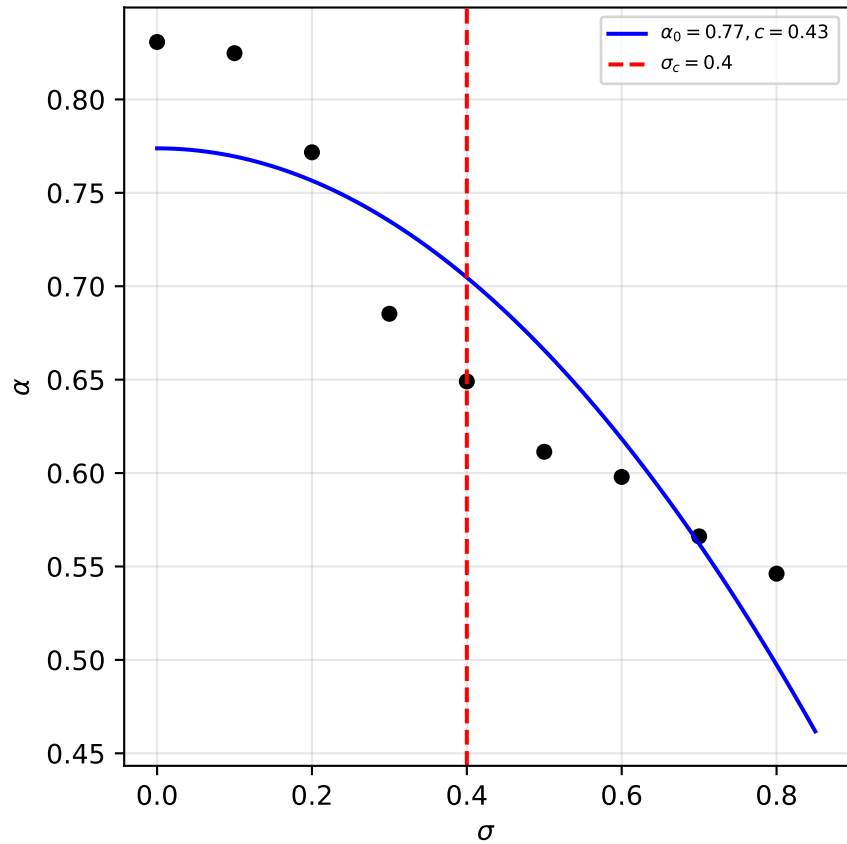
(a) 1D Scaling $\alpha = 0.858$



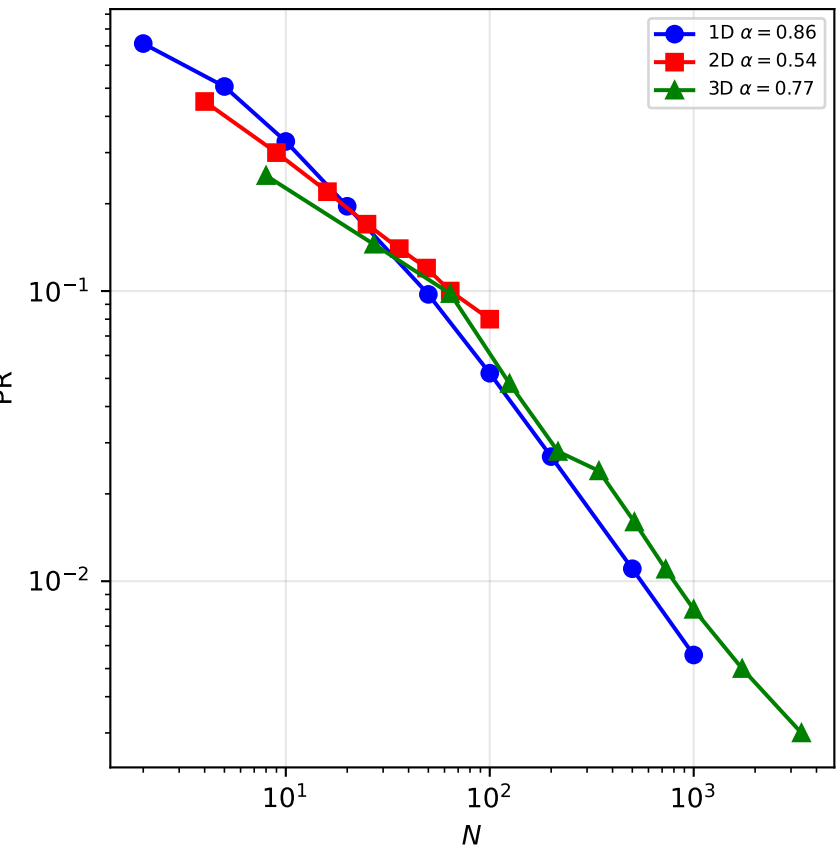
(b) Energy Sensitivity
(range 19.2%, $N = 20$)



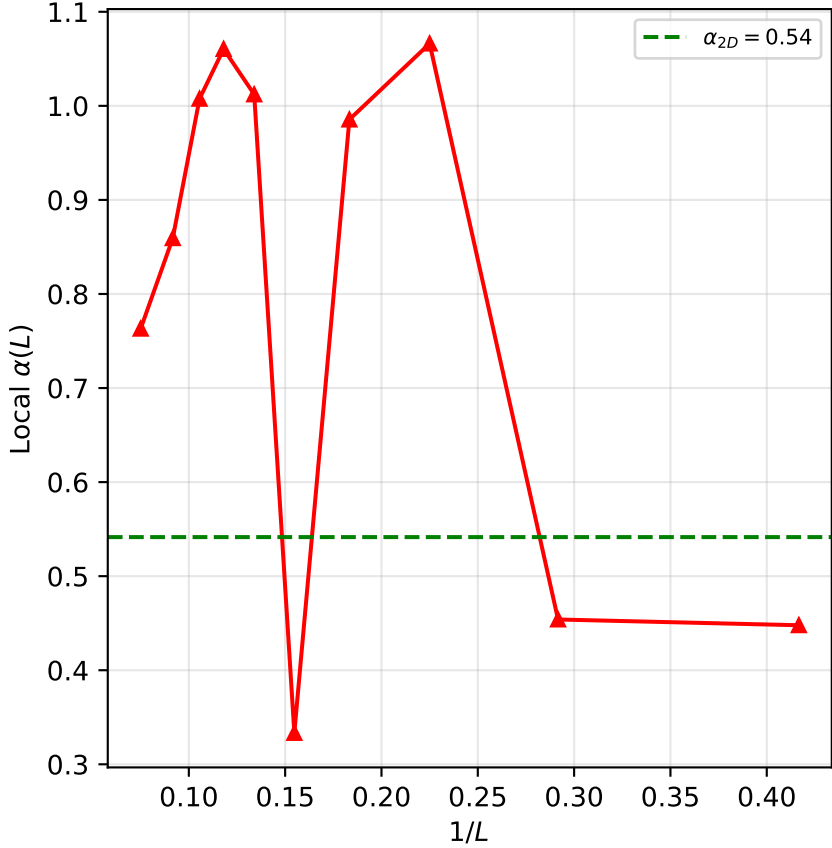
(c) Disorder Dependence



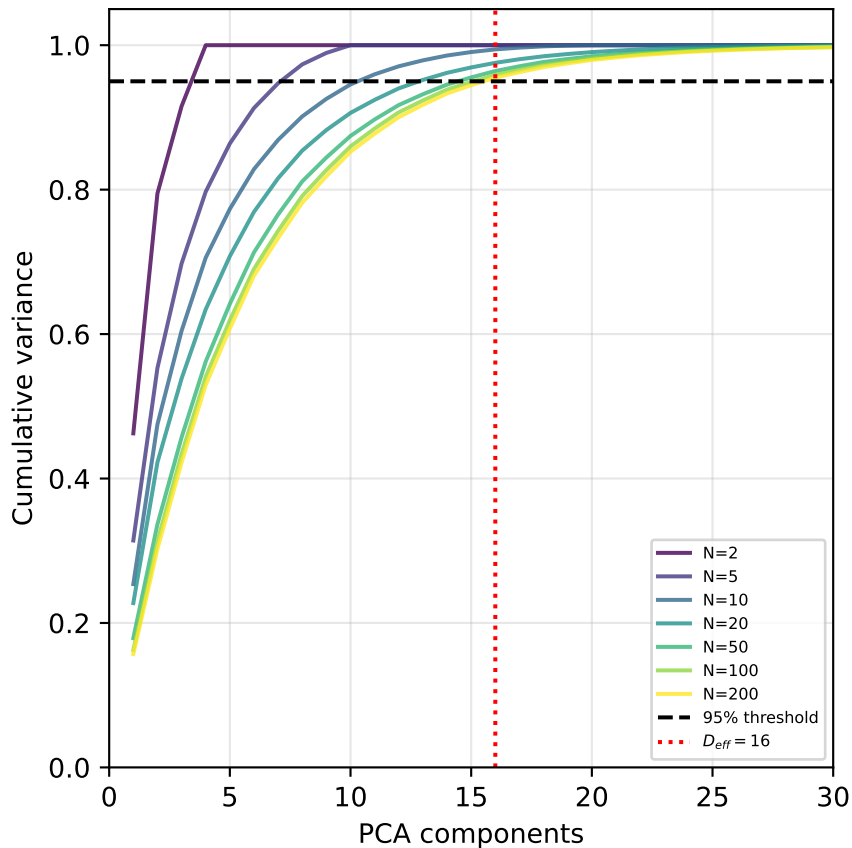
(d) All Dimensions



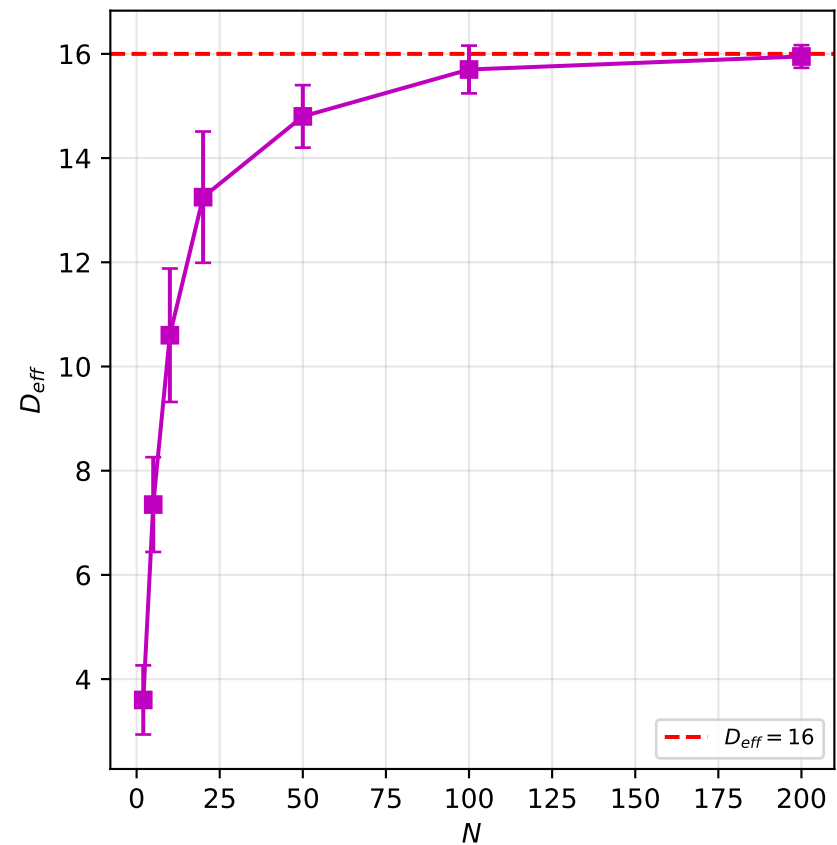
(e) 3D Finite-Size Crossover



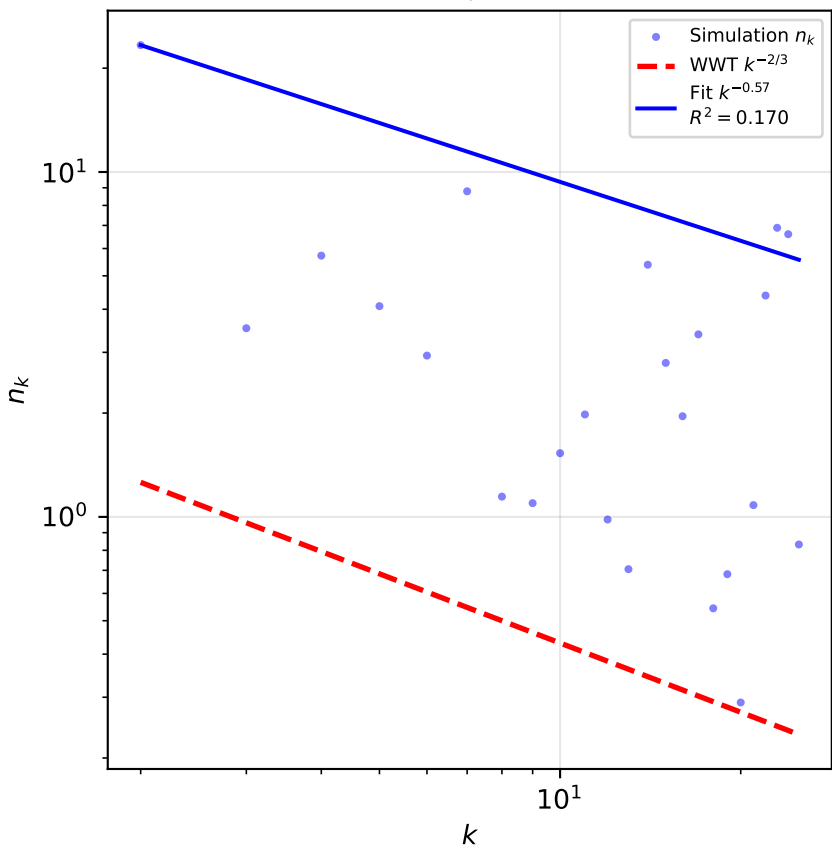
(f) D_{eff} Saturation



(g) D_{eff} vs N



(h) KZ Spectrum



(i) Summary

Metric	Value
α_{1D}	0.858 ± 0.06
α_{2D}	0.542 ± 0.05
α_{3D}	0.767 ± 0.05
D_{eff} saturation	~ 16
Energy robustness	19.2% range
σ_c	≈ 0.4
KZ exponent	-0.567
KZ R^2	0.1700
CV error	0.19%